

Research papers

Soil physicochemical properties and crusts regulate the soil infiltration capacity after land-use conversions from farmlands in semiarid areas

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ARTICLE INFO

This manuscript was handled by Yuefei Huang, Editor-in-Chief, with the assistance of Thomas Wöhling, Associate Editor

Keywords:

Double-ring infiltrometer
The Loess Plateau
Plough pan
Soil moisture replenishment
Soil structure
Vegetation restoration

ABSTRACT

Large scale land-use conversions played important role in regulating the process of soil infiltration, and thus changing soil moisture recharge. However, how the soil properties influence infiltration after farmland conversions remain unclear in the semiarid areas. This study used a double-ring infiltrometer to conduct infiltration experiments in soil surface and subsurface among land-uses on the Loess Plateau, to explore the mechanism of soil physicochemical properties and crusts on infiltration. The results showed that the revegetation lands effectively improved infiltration properties, while no significant differences were observed between the orchards and the farmlands. The loose soil structure, stable aggregates, higher organic matter and root mass density improved while soil crusts impeded infiltration capacity after farmland conversions. The impeding effect of the biocrusts presented throughout the infiltration process while physical crusts impeded infiltration only at the initial stage. Soil physical properties (i.e., bulk density, porosity, macroaggregate, texture, moisture), biochemical properties (i.e., soil organic matter, roots) and crusts were major factors in the variation of the surface (0–20 cm) infiltration properties, while subsurface (20–40 cm) infiltration were mainly influenced by soil physical properties. The deeper soils had a greater impact on steady infiltration rate than initial infiltration rate. Due to the plough pan in agricultural lands, the subsurface soil (20–40 cm) had a strong effect (42.1%) on the steady infiltration rate of the surface (0–20 cm) soil, even close to the surface (0–20 cm) soils (49.8%). These findings improve the comprehensive understanding of the soil hydrological processes after farmlands conversions in semiarid areas.

1. Introduction

Soil water availability constrains the effectiveness of vegetation restoration (Jian et al., 2015). Rainfall is the most important factor influencing species occurrence and diversity (Fu et al., 2011). Soil infiltration after rainfall, an essential hydrological process in terrestrial ecosystems, represents the most important source of soil water recharge regionally (Bormann and Klaassen, 2008). Plants participate in the terrestrial ecosystem water cycle by acquiring, transporting and transpiring soil water. Land-use conversion accompany with vegetation cover changes is one mainly factor to influence on soil infiltration by

changing soil properties and evapotranspiration (Liao et al., 2021a). Many studies have shown that appropriate soil structure can effectively increase the soil infiltration rates and the efficiency of water recharge (Bormann and Klaassen, 2008; Leung et al., 2015; Wu et al., 2019). Therefore, soil infiltration capacity will change significantly with land-use conversions.

Farmland conversions could greatly change the soil properties, which can be attributable to two main factors. The first is the change in human disturbances, particularly tillage (Wu et al., 2023). The second is the influence of different vegetation, such as root activity influencing soil structure, canopy cover changing the distribution of rainwater and

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<https://doi.org/10.1016/j.jhydrol.2023.130283>

Received 18 April 2023; Received in revised form 12 September 2023; Accepted 16 September 2023

Available online 7 October 2023

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litter inputs altering soil nutrients (Jian et al., 2015; Cui et al., 2022b, Dong et al., 2022). Under cultivation, the agricultural soils have experienced dramatic changes, especially in the surface (plow layer: 0–20 cm) and subsurface (plough pan: 20–40 cm), the latter often exhibit significant traits of reducing soil porosity and increasing soil bulk density after compaction (Verbist et al., 2007; Li et al., 2019; Wang & Zhang, 2017). There is a consensus that vegetation restoration will significantly alter soil physicochemical properties, such as bulk density, porosity, organic matter, moisture and texture (Deng et al., 2016; Bai et al., 2020; Yi et al., 2021). These changes in soil properties will greatly affect the soil infiltration and water recharge efficiency (Beven and Germann, 2013; Chen et al., 2013; Chandler et al., 2018; Qiu et al., 2022). Hence, it is important to explore the changes in soil properties after farmland conversions and their effect on infiltration, especially in the surface and subsurface soil. In addition, infiltration is the property of whole soil profile and soils at different depths have more or less influence on infiltration. How soils at different depths affect the changes in soil infiltration is also worth investigating after land-use conversions, especially in the comparison between agricultural and revegetation lands. However, current studies are more focused on the relationships between soil properties and infiltration properties by considering the soil profile as a whole.

Soil crusts are widespread and profoundly affect surface hydrological processes in arid and semi-arid regions. Due to different formation mechanisms, soil crusts can be divided into physical crusts and biocrusts. Physical crusts are fine particulate layers that form on the surface under the influence of splash, slaking, swelling, sedimentation and drying (ISSS, 1996), reducing soil permeability by sealing the surface (Nciizah and Wakindiki, 2015). Biocrusts are mixture of the uppermost few centimeters of the soil surface and symbiotic photoautotrophic communities (Kidron et al., 2022). The effect of biocrusts on soil infiltration remains controversial, such as improvement (Jiang et al., 2018; Liu et al., 2019), impediment (Wang et al., 2017) or non-effect (Kidron and Budel, 2014). Physical crusts are a common feature of most cultivated soils (Nciizah and Wakindiki, 2015), while biocrusts may present in grasslands (Sun et al., 2022), shrublands (Liu et al., 2019) and forestlands (Chen et al., 2020). With the conversions of land-use, the distribution and properties of soil crusts will inevitably change, and thus leading to a variation in soil infiltration.

Many methods were used to measure soil infiltration, such as single-ring infiltrometer (Liu et al., 2019), double-ring infiltrometer (Zhang et al., 2017), disc permeameter (Chandler et al., 2018), Hood infiltrometer (Patra et al., 2023), rainfall simulators (Comino et al., 2016), etc. The double-ring infiltrometer is one of the most classic and commonly used method for measuring soil infiltration properties in the field, and is often used as a basis for judging the results of other methods (Angulo-Jaramillo et al., 2000; Cui et al., 2022a). Through double-ring infiltration experiments, Wu et al. (2021) found that decaying roots after revegetation promoted soil infiltration; Qiu et al. (2023) found that revegetation had a positive impact on the saturated hydraulic conductivity and preferential flow; and Cui et al. (2022a) found that short-term forestlands-to-farmland conversion increased infiltration performance and thus alleviated soil water deficits. However, these current studies have focused on the infiltration properties of the soil surface and lack the study of the subsurface. In addition, more reports have investigated the effect on infiltration rates and infiltration volumes, without further exploring the distribution of infiltrated water in the vertical profile.

Large-scale land-use conversions from farmlands have occurred on the Loess Plateau, where serious soil erosion is faced (Deng and Shangguan, 2021). Soil erosion has led to serious ecological problems and land degradation, the latter becoming an important factor in the poverty of local farmers at certain times in history. The Chinese government launched the “Grain for green” Program in 1999 to control soil erosion and improve the ecological environment (Deng et al., 2014). Through the conversion of farmlands into forestlands, shrublands and grasslands, the vegetation cover of the Loess Plateau has increased by

~34% from 1999 to 2021 (Deng and Shangguan, 2021). In addition, plenty of farmlands has been converted into apple orchards, organized by the local government, in order to increase the income of farmers. Currently, there are over 620,000 ha of apple orchards on the Loess Plateau, contributing a quarter of China’s apple production (Shaanxi Provincial Bureau of Statistics, 2022). Both the conversion of farmlands to revegetation lands and apple orchards will result in changes in soil physicochemical properties and soil crusts. During infiltration, soil crusts will first affect water migration, and when water passes through the thin (usually a few millimeters thick) soil crusts, the soils under the crusts will also be involved in the regulation of water. Therefore, upper soil crusts and the physicochemical properties of the underlying soil will co-affect the infiltration process. However, to our knowledge, few studies have reported both the effects of soil physicochemical properties and soil crusts on infiltration.

In order to fill this knowledge gap, we carried out infiltration experiments on the Loess Plateau, determined the soil physicochemical properties and investigated the properties of soil crusts. The primary objectives of this study were to explore the changes of soil infiltration capacity after farmland conversions, and to clarify its regulation mechanism. The novelties may include two aspects: (1) integrated consideration of the effects of changes in soil physicochemical properties and crusts on soil infiltration after land-use conversions; (2) quantified the contribution of soils at different depths to the changes in infiltration after land-use conversions. This study will help to improve our understanding of soil hydrological processes after farmland conversions in semi-arid areas.

2. Materials and methods

2.1. Study areas

The study site is located in Yan’an city, Shaanxi province, China (36°11′–37°09′N, 109°21′–110°03′E), belonging to a semi-arid climate and a temperate continental monsoon climate. The mean annual rainfall is around 400–600 mm, with more than 60% occurring between July and September. The mean annual temperature is 9.4° and the annual frost-free period is 170–186 days. The soil is classified as a silty loess, mainly developed from the parent body, and is characterized by being homogeneous, porous, friable, yellowish or light brown. Historically, crops have been grown on a large scale in the district. In 1999, plenty of farmlands were converted to forestlands, shrublands and grasslands in response to the policy of “Grain for Green” Program. In addition, the area is a suitable area for apples (Liao et al., 2021a), and many farmlands were also converted into orchards in order to increase the income of farmers.

2.2. Experimental design

Five land-use types were selected for the experiment: forestlands, shrublands, grasslands, orchards and farmlands (Fig. 1). The maximum distance between the different site types is no more than 5 km, with a difference in altitude ranging from 1229 to 1297 m. Six plots (20 m × 20 m) were randomly established for each land-use type. The forestlands, shrublands and grasslands were all converted from farmlands in response to the “Grain for Green” with a 23-year restoration period. The dominant species of forestlands and shrublands are artificially planted *Robinia pseudoacacia* and *Hippophae rhamnoides*, respectively, and the grasslands were naturally restored with *Artemisia scoparia* and *Incarvillea sinensis* as the dominant species. The apple (*Malus pumila*) orchards were converted from farmlands about 30 years ago. The primary farmland species was *Zea mays*. Orchards and farmlands are cultivated once a year with small rotary tillers, with a tillage depth of about 20 cm. More basic information about the sample sites is shown in Table 2.

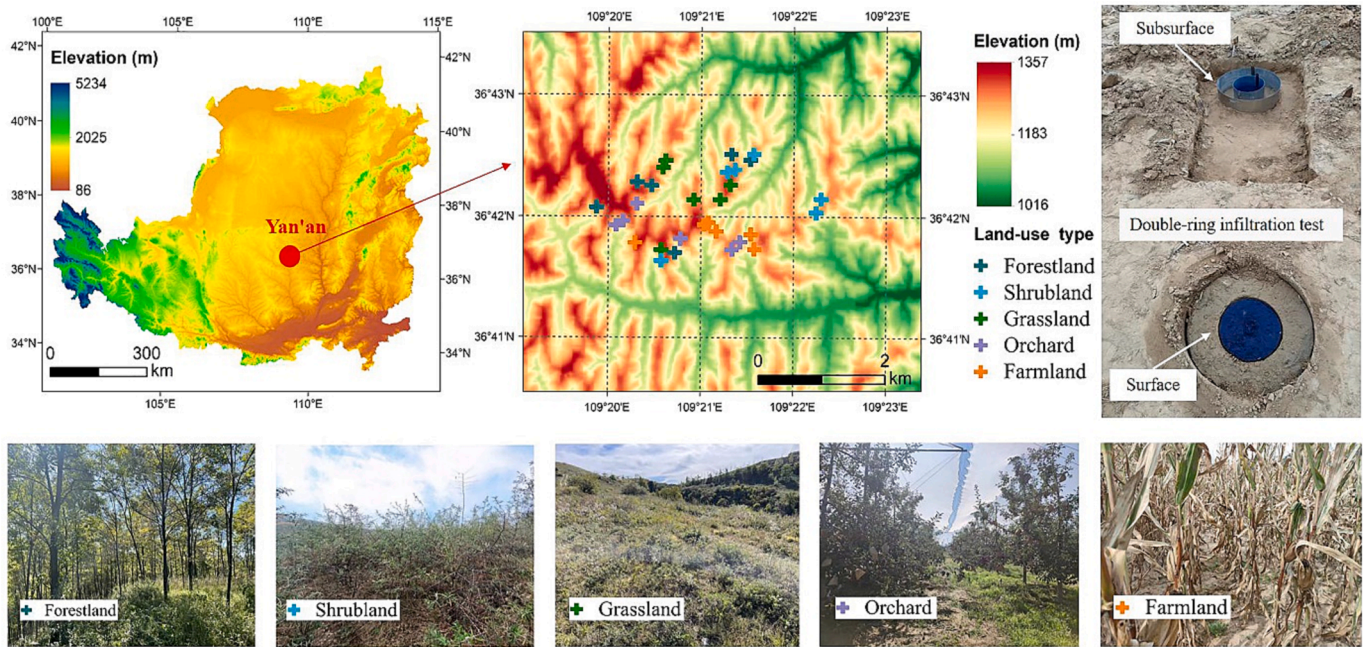


Fig. 1. The information and photos of the study area and experimental treatments.

2.3. Measurement of infiltration properties

Soil infiltration properties were measured using a double-ring infiltrometer with an inner and outer diameter of 25 cm and 50 cm respectively. Infiltration experiments were carried out in each plot, which meant that each land-use type was repeated six times. Two infiltration tests, surface and subsurface, were carried out in each plot, with the subsurface infiltration experiment was carried out at a distance of about 2 m from the surface infiltration experiment, while the top 0–20 cm of soil was removed with a shovel before the experiment (Fig. 1).

After the removal of the herbs and litters, the double-ring infiltrometer was gently and vertically inserted 10 cm into the soil to avoid, as far as possible, any damage to the soil structure. Methylene blue solution and water were added rapidly and simultaneously to the inner and outer ring respectively, stopped when the water level reached 4 cm. Soil infiltration process measured by the falling head method (Zhang et al., 2017). The time at which the water level in the inner ring dropped by 5 mm was recorded and when the water level dropped to 1 cm, we refilled it to 4 cm, which means that 3 cm of water was added each time. When three consecutive measurement times remain almost constant, we assume it to have reached the stable infiltration stage (Rezaei et al., 2016; Zhang et al., 2017). Throughout the infiltration process, the water level in the outer ring needed to be equal to that in the inner ring.

Guo et al. (2019) and Qiu et al. (2023) defined the initial infiltration rate (IIR, list of abbreviations were shown in Table 1) as the mean value of the first 3 min (0.05 h), while Cui et al. (2022a) defined the first 8 min to be the initial infiltration stage. By fitting the rate of change of infiltration rate with infiltration time, we found that the end of the third minute (0.05 h) was a point in time when the change rate infiltration rate changed from a rapid to a slow change phase (Fig. S1). Therefore, we took the mean infiltration rate in the first 3 min as the IIR. The stable infiltration rate (SIR) was defined as the mean value of the last three infiltration rates (Guo et al., 2019; Wu et al., 2021).

Notably, in order to study the replenishment of infiltrated water on the vertical profile of the soil for a fixed amount of infiltration, we set the end conditions of the experiment to be:

Table 1

List of abbreviations in this paper.

Abbreviations	Full name	Abbreviations	Full name
Soil properties		Infiltration properties	
SWC	Soil water content	IIR	Initial infiltration rate
BD	Soil bulk density	SIR	Stable infiltration rate
FC	Field capacity	SMR	Soil moisture replenishment
Macro	Macroaggregate	ID	Infiltration depth
L-micro	Large microaggregate	Measurement methods and statistical analysis	
S-micro	Small microaggregate	WDPT	Water droplet penetration time test
SOM	Soil organic matter	MED	Molar of an ethanol droplet test
RMD	Root mass density	ANOVA	Analysis of Variance
PC	Physical crust cover	RDA	Redundancy analysis
BC	Biocrust cover	VPA	Variance partitioning analysis
SWR	Soil water repellency	MLRA	Multivariate linear regression analysis

- (1) at the end of the 4th water addition to the inner ring, i.e., a total infiltration amount of: $40 + 3 \times 30 = 130$ mm.
- (2) if the infiltration time has exceeded 2 h before the 4th water addition, we stop the measurement but still add water to ensure the same amount of infiltration (130 mm).

After the infiltration experiments, the rings were removed and the water was allowed to infiltrate fully. Shade nets were covered over the wetted areas to reduce water evaporation. Then, after 24 h, soil samples were collected at 5 soil layers (20 cm intervals over the 100 cm-depth) near the center of the inner circle using a 25 mm diameter soil auger drill. Three samples were collected at each soil layer and mixed for subsequent determination of soil water content (SWC) after infiltration. Then, at the position of the center line of the inner ring, a vertical soil profile was excavated and the depth of the wetted front was measured using a steel ruler. Before infiltration, soil samples were collected in each plot using the same method near the location where the infiltration rings

will be installed to determine the SWC before infiltration.

Infiltration is the property of whole soil profile. For ease of description, the infiltration properties measured by the surface infiltration experiment were defined as the surface infiltration properties, and the infiltration properties measured by the subsurface infiltration experiment were defined as the subsurface infiltration properties.

2.4. Soil sampling and analysis

Three 1 × 1 m sample quadrats were randomly set up in each plot. Firstly, vegetation community characteristics were investigated, dominant species were identified, and aboveground biomass and litter biomass were collected and measured. In forestlands, shrublands and apple orchards, the above-ground biomass of *Robinia pseudoacacia*, *Hippophae rhamnoides* and *Malus pumila* were calculated using biomass allometric equation based on trunk diameter (diameter at breast height and ground diameter) and plant height (Liu and Liu, 1994; Liu et al., 2011; Lyu et al., 2021). Then, soil samples at 5 soil layers (20 cm intervals over the 100 cm-depth) were collected in each square using a 9 cm diameter root drilling, and soil samples from the same soil layer were mixed into a composite sample, resulting in 150 soil samples (5 land-use types × 6 plots × 5 soil layers). A cutting ring (100 cm³ in volume) and stainless steel boxes (2000 cm³ in volume) were used to collect undisturbed soil from each soil layer for subsequent analysis of soil bulk density (BD), porosity, field capacity (FC) and aggregates.

BD, porosity and FC were measured using the cutting ring method (Bao, 2007). Soil organic matter (SOM) was determined using dichromate oxidation (Dong et al., 2022). Soil pH was determined using a pH meter with a soil to water ratio of 2.5:1. The mixture samples of soil and root collected by root drilling were used to determine root mass density (RMD) using water washing and oven-dried method. Soil texture was determined using a Mastersizer 2000 laser particle sizer (Malvern Instruments Ltd., Malvern, UK). The aggregate size distribution was measured by wet sieving methods (Dong et al., 2022), and the aggregate size fractions obtained were divided into > 0.25, 0.25–0.053, and <0.053 mm, representing macroaggregate (Macro), large microaggregate (L-micro), and small microaggregate (S-micro), respectively.

The SWC of soil samples is determined by drying to constant weight using an oven. The amount of soil moisture replenishment (SMR) on the vertical profiles was calculated using the following formula:

$$SMR = SWC_1 \times h \times BD - SWC_0 \times h \times BD$$

Here, SWC_1 is soil water content of a specific soil layer after infiltration; SWC_0 is the soil water content of a specific soil layer before infiltration; h is the depth of a specific soil layer (mm); BD is the bulk density of a specific soil layer (g cm⁻³).

2.5. Measurements of soil crusts properties

In our research, the physical crusts were found in orchards and farmlands and the biocrusts were found in orchards (moss crusts) and grasslands (cyanobacteria and moss mixed crusts). At the location where the double-ring infiltrometer will be installed, we took digital photographs of the ground after we had removed the herbs and litters. Then, image analysis was carried out by Image J software (National Institutes of Health) to calculate the crust cover based on the number of crust pixels as a proportion of the total number of pixels, further details are provided in Sun et al. (2022). 10 cm × 10 cm × 2 cm Petri dishes were used to collect 100 cm² of random samples of soil crust and bare soil from each plot, and further processed in the laboratory to measure soil crust thickness, soil texture and SOM. In addition, cutting rings (2 cm in height and 60 cm³ in volume) were used to sample a 2 cm surface layer of soil to measure BD, porosity, SWC and FC. Crust thickness was measured using a digital caliper, and the biomass of biocrusts was separated from soil with water through a 2 mm sieve and dried at 65 °C for 24 h for weighing. BD, porosity, FC, SWC, SOM and texture were

measured by referring to section 2.4.

The water droplet penetration time (WDPT) (Dekker et al., 2009) and the molar of an ethanol droplet (MED) test (Roy and McGill, 2002) are the two most commonly used methods to measure soil water repellency (SWR). After the removal of the herbs and litters, we carried out WDPT (6 repetitions) and MED testing (3 repetitions) in each plot. Three drops of water were continuously placed to the bare soil and soil crusts surface using a standard rubber-tipped dropper and the time for the last droplet to completely infiltrate into the soil was recorded (Chandler et al., 2018), this is WDPT. During the MED test, seven different concentrations of ethanol solutions (0, 3%, 5%, 8.5%, 13%, 24% and 36%) were used in the test, with the lowest ethanol concentration at which at least 3 out of 5 droplets penetrated the soil surface within 5 s, which determined the repellency class (Papierowska et al., 2018).

2.6. Statistical analysis

In this study, we divided the 0–100 cm soil layer into three soil layers to correspond to the surface, subsurface and the deep soil respectively (0–20 cm, 20–40 cm, 40–100 cm). The differences of the vegetation properties, soil biocrust properties and infiltration properties among the different land-uses were analyzed by the one-way analysis of variance (ANOVA). Two-way analysis of variance (ANOVA) was performed to evaluate the effects of land-use type and soil depth on soil physicochemical properties. To assess the level of SWR under both methods, we used the standard SWR assessment scale (Table 3) proposed by Papierowska et al. (2018). Redundancy analysis (RDA) was used to identify the relationship between response variables (infiltration properties) and explanatory variables (soil properties of the corresponding soil layer), and Monte Carlo permutation test was used to verify the correlation between the response variables and the explanatory variables. Variance partitioning analysis (VPA) was performed to quantify the relative contribution of explanatory variables to the variation in response variables. RDA and VPA were performed by the R package “vegan”. Multivariate linear regression analysis (MLRA) was carried out using SPSS Statistics, ver. 26.0 (SPSS Inc., Chicago, IL, USA) to analyze the contribution of soil properties of different soil layers to the variations of infiltration rates. More information on the data and processing methods was provided in the analytical framework (Fig. S2).

3. Results

3.1. Changes in soil physicochemical properties and soil crusts since farmland conversions

Compared to farmlands, higher BD and lower porosity and FC at 0–40 cm were found in revegetation lands ($p < 0.05$, Fig. 2). After land-use conversions, all treatments promoted the formation of Macro (0–100 cm) and had lower SWC at 20–100 cm soil ($p < 0.05$, Fig. 2, Fig. S3). As for texture, sand content was lower at 0–40 cm in forestlands and shrublands ($p < 0.05$), and orchards had higher sand and lower silt content at 0–20 cm soil ($p < 0.05$, Fig. 2). Distinct subsurface compaction (plough pan) occurred in orchards and farmlands, with BD first increasing and then decreasing with soil depth, while porosity and FC first decrease and then increase. Compared to farmlands, the SOM was higher at 0–40 cm in the revegetation lands ($p < 0.05$), while no significant difference occurred in orchards. Forestlands and shrublands had higher RMD at 0–100 cm ($p < 0.05$), whereas grasslands higher only at 0–20 cm and lower at 40–100 cm soil ($p < 0.05$). The soil pH at 0–100 cm was lower in forestlands ($p < 0.05$), while the remaining treatments did not show significant differences.

Soil biocrusts occupied 60.5% and 47.2% of the area in grasslands and orchards respectively, while physical crusts occupied 16.5% and 73.9% of the area in orchards and farmlands, respectively (Table 4). The thickness of the biocrusts were greater than that of the physical crusts ($p < 0.05$). Compared to bare soil, physical crusts had higher BD and lower

Table 2

Basic information of sampling sites.

Land use types	Slope (°)	Coverage (%)	Aboveground biomass (g m ⁻²)	Litter biomass (g m ⁻²)	Tillage frequency (times year ⁻¹)	Shannon-Wiener Index	Dominant species
Forestlands	3–5	75.8 ± 11.1a	6107.8 ± 1793.6a	213.8 ± 45.6a	–	1.8 ± 0.4a	<i>Robinia pseudoacacia</i>
Shrublands	3–5	57.5 ± 12.9b	3988.9 ± 1417.3b	54.9 ± 12.7b	–	1.7 ± 0.7a	<i>Hippophae rhamnoides</i>
Grasslands	6–9	58.3 ± 9.8b	129.1 ± 31.1d	43.9 ± 17.1b	–	1.6 ± 0.5a	<i>Artemisia scoparia</i> , <i>Incarvillea sinensis</i>
Orchards	<3	50.8 ± 18.6b	2624.7 ± 390.3c	12.9 ± 7.2c	1	0.9 ± 0.5b	<i>Malus pumila</i>
Farmlands	<3	–	2201.1 ± 464.6c	10.7 ± 5.1c	1	–	<i>Zea mays</i>

Note: Lowercase letters denote differences in different land-use type under the same soil crust type at $p = 0.05$ level.

Table 3

Standard soil water repellency assessment scale.

Class	Descriptive label	Water drop penetration time (s)	Molarity of an ethanol droplet (%)
1	Hydrophilic (wetttable)	<5	0 and 3
2	Slightly hydrophobic	5–60	5
3	Moderately hydrophobic	60–600	8.5
4	Strongly and very strongly hydrophobic	600–3600	13 and 24
5	Extremely hydrophobic	≥3600	36

porosity and FC ($p < 0.05$), but did not show significant differences on SWC, texture and SOM. The SOM was higher than bare soil in biocrusts ($p < 0.05$). In addition, lower BD and sand content, higher TP and FC were found in the biocrusts in the grasslands, and biocrusts in orchards had lower sand and higher clay content ($p < 0.05$). Compared with bare soil, soil crusts had higher SWR levels, especially biocrusts (Fig. 3). According to the results of the WDPT test, more soil in the physical crusts showed slightly hydrophobic, and most soils in the biocrusts showed moderately hydrophobic. The results of the MED test were similar to those of the WDPT test, while with a stronger degree of SWR, e.g., in the case of biocrust soils, part of the soils showed strongly hydrophobic.

3.2. Changes in soil infiltration properties since farmland conversions

Before reaching the stop conditions we set, the infiltration test for each plot reached stable infiltration stage. In general, with increasing infiltration time, the soil infiltration rate and the absolute values of the change rate of soil infiltration rate decreased (Fig. 4, Fig. S1). Infiltration experiments, both in the surface and the subsurface, showed that the revegetation lands had higher IIR and SIR, and reached significant levels in forestlands and shrublands ($p < 0.05$). In the surface, compared with farmlands, the IIR and SIR of forestlands, shrublands and grasslands were higher by 67.3%, 81.6%, 5.3% and 177.2%, 153.1%, 47.6%, while those of orchards were lower by 32.5% and 30.5%. And in the subsurface, IIR and SIR of forestlands, shrublands, grasslands and orchards were 217.5%, 165.1%, 105.6%, 61.9% and 195.7%, 171.5%, 105.5%, 64.8% higher than farmlands respectively. Between the two soil layers, only the IIR in the surface was greater than that in the subsurface in farmlands ($p < 0.05$), while the IIR in the remaining treatments did not differ significantly (Fig. 4). In addition, the SIR in the forestlands, shrublands and farmlands were greater in the surface than in the subsurface ($p < 0.05$). The opposite trend was observed in the orchards, and no significant differences were found in the grasslands.

For the same amount of infiltration (130 mm), compared to farmlands, forestlands, shrublands and grasslands had higher SMR in the vertical profiles ($p < 0.05$), by 17.9%, 23.1% and 14.4% in the surface and 27.7%, 33.5% and 40.4% in the subsurface (Fig. 5). The revegetation lands had higher infiltration depth (ID) compared to the two agricultural lands (Fig. 5a–b). This can be also confirmed by the distribution of SMR in the individual soil layers (Fig. 5c–d). In the 0–40 cm soil layers,

the difference of SMR between revegetation lands and agricultural lands was small, while SMR in deeper layers (40–100 cm) was lower in the agricultural lands ($p < 0.05$). In terms of the proportion of SMR in each soil layer, a greater proportion of the agricultural lands was distributed in shallow soil layers and less in deeper soil layers than in the revegetation lands.

3.3. Factors influencing soil infiltration

In the experiments conducted in the surface and subsurface, RDA1 and RDA2 explained 56.3%, 6.3% and 37.5%, 3.8% of the variance of the response variables respectively (Fig. 6a–b). Monte Carlo permutation test (Table S1) showed that in the surface, porosity, FC, SOM, RMD, silt and clay content were significantly positively correlated with the infiltration properties; the BD, L-micro, S-micro, physical crust cover (PC) and biocrust cover (BC) were significantly negatively correlated with the infiltration properties. In the subsurface, without the influence of soil crusts, porosity, FC, silt and clay content were significantly positively correlated with the infiltration properties, and the BD, L-micro, S-micro and SWC were significantly negatively correlated with the infiltration properties.

In the surface, soil physical properties, biochemical properties and soil crusts explained 65.0%, 54.6% and 69.8% of the response variables respectively, and 37.1% of the changes could not be explained (Fig. 6c–d). The majority of the variation is explained by all three together, and only the soil crusts have a separately explained component (9.4%). In the subsurface, soil physical properties and biochemical properties explained 47.9% and 5.1% of the variation, respectively, with a joint explanation of 3.2% and residuals of 50.2%. VPA showed that all three parts are major factors in the variation of the infiltration properties in the surface, while the physical properties are the main factor in the variation of the infiltration properties in the subsurface.

The contributions of soil properties of different soil layers could explain 72.8%, 79.4% of IIR and SIR in the surface (Fig. 7a–b) and 53.4%, 54.3% in the subsurface (Fig. 7c–d). Compared to the IIR, the deeper soils have a greater impact on the SIR. The relative effects of soil properties of 0–20 cm, 20–40 cm, 40–100 cm soil layer on surface IIR were 55.1%, 29.2%, 9.4%, and on the SIR were 49.8%, 42.1%, 8.1%, respectively. As for the subsurface, the relative effects of soil properties of 20–40 cm and 40–100 cm soil layers on the IIR were 77.1%, 22.9%, and on the SIR were 69.5% and 30.6%, respectively.

4. Discussions

4.1. Effects of land-use conversions on soil physicochemical properties and crusts

Soil properties are the main factors that influence infiltration capacity following land-use conversions (Sun et al., 2018). Both human activities and plant-soil interactions can cause significant changes in soil properties, although there may be undeniable interference from spatial

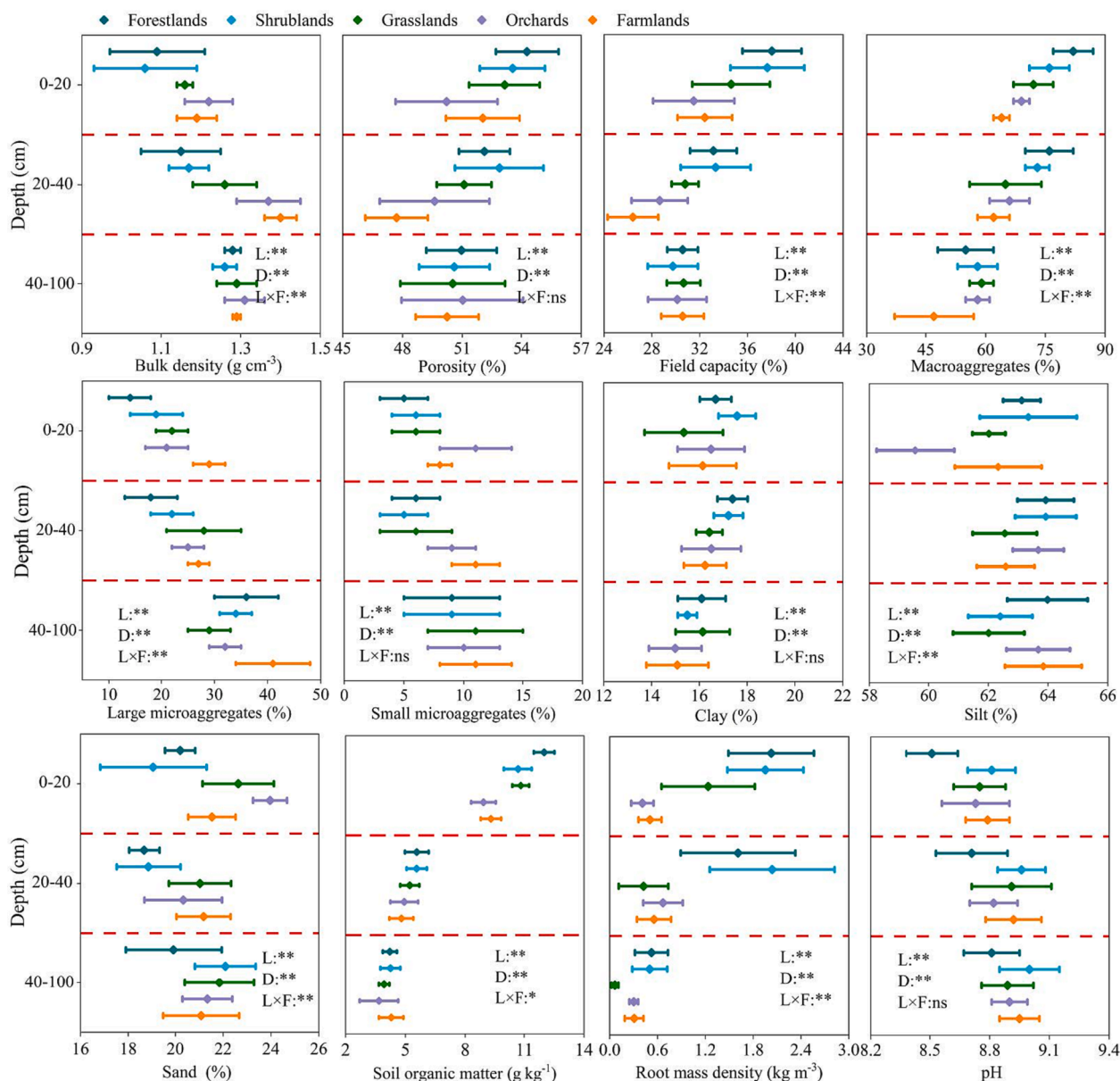


Fig. 2. Vertical variation of soil physicochemical properties under different lands uses. Data are means $\pm$ standard error (SE). D: depth, L: land-use type, D $\times$ L: the interaction between depth and land-use type. * $p < 0.05$, ** $p < 0.01$, ns denotes not significant ($p > 0.05$).

variability, which can cause changes in soil properties on scales of a few metres (Marzvan et al., 2021). Subsurface soil compaction has been identified as an important cause of soil degradation (Verbist et al., 2007), which is found in both orchards and farmlands in our study (Fig. 2). Due to the absence of tillage, the subsurface soil did not show compaction under the revegetation lands. Compared with farmlands, forestlands and shrublands had higher above-ground biomass, litter biomass and root biomass ($p < 0.05$), while grasslands had higher litter biomass and root biomass of 0–20 cm soil ($p < 0.05$; Table 2, Fig. 2). The increase in biomass will lead to changes in SWC and SOM, due to the changes in leaf interception, root water uptake (Jia et al., 2017; Jian et al., 2015) and input of organic materials (Aubinet et al., 2009; Wiesner et al., 2022). Soil organic cementing matter and the absence of tillage will also promote the formation of soil macroaggregates and changes in texture (Zhang et al., 2013; Okolo et al., 2020). It has been found that the mechanical action of the root system (penetration, adhesion and stripping) and the root exudates caused the breakdown of minerals increase the content of fine particles (clay and silt) and

macroaggregate (Demenois et al., 2018; Zhao et al., 2022). Furthermore, root growth and decay due to vegetation restoration also improved pore structure and inter-pore connectivity in the soil (Bowker et al., 2013; Wu et al., 2021).

Soil physical crusts are widespread in cultivated lands. At lower vegetation and litter cover, the soil aggregates break down into finer fragments and redistribute under tillage and raindrop impact. These fine fragments will then redistribute vertically within the top few millimeters by infiltrating water and form a physical crust after drying and consolidation (Le Bissonnais et al., 2005; Nciizah and Wakindiki, 2015). de Jong et al. (2011) emphasized that physical crusts increased BD and reduces porosity, which is consistent with our results. The biocrust is widely regarded as an ecosystem engineer due to its strong influence on the properties of the topsoil (Chamizo et al., 2012; Kakeh et al., 2018). Some studies found that the biocrusts reduced BD and increased soil porosity, SOM and fine particles (clay and silt) (Sun et al., 2022; Li et al., 2022). In our study, a similar phenomenon occurred in grasslands, while biocrusts in orchards only increased the SOM and fine particle and had

Table 4

Properties in surface soil (0–2 cm depth) of bare soil and soil crusts.

Soil crust type	Land-use type	Crust cover (%)	Crust thickness (mm)	Total biomass (g/m ²)	Soil water content (%)	Bulk density (g cm ⁻³)	Porosity (%)	Field capacity (%)	Clay content (%)	Silt content (%)	Sand content (%)	Soil organic matter (g kg ⁻¹)
Bare soil	Forestlands	–	–	–	98.6 ± 7.4a	0.99 ± 0.11bc	58.9 ± 2.5b	39.6 ± 2.4a	15.7 ± 1.9ab	66.3 ± 3.1a	18.0 ± 1.6bc	16.0 ± 1.1b
	Shrublands	–	–	–	104.8 ± 27.8a	0.93 ± 0.03c	65.0 ± 5.8a	38.6 ± 2.9a	16.9 ± 1.3a	66.2 ± 3.7a	16.9 ± 2.8c	17.6 ± 0.6a
	Grasslands	–	–	–	117.8 ± 15.9aA	1.09 ± 0.09abA	55.0 ± 6.2bcB	34.6 ± 2.5bB	15.5 ± 2.0abA	64.8 ± 2.9abA	19.6 ± 1.6abA	16.6 ± 1.6abB
	Orchards	–	–	–	118.0 ± 16.3aA	1.19 ± 0.07aB	52.6 ± 4.1cA	32.4 ± 1.5bA	15.4 ± 1.3abB	63.2 ± 1.8bA	21.5 ± 1.2aA	11.3 ± 0.8cB
	farmlands	–	–	–	110.7 ± 26.3aA	1.15 ± 0.05aB	50.8 ± 3.2cA	31.5 ± 2.5bA	15.2 ± 0.9bA	65.3 ± 0.5abA	19.5 ± 1.1bA	11.2 ± 0.8cA
Physical crust	Orchards	16.5 ± 9.9b	5.1 ± 0.7a	–	113.7 ± 20.5aA	1.29 ± 0.06aA	46.9 ± 3.7aB	27.5 ± 1.6aB	15.4 ± 0.8aB	64.5 ± 2.7aA	20.1 ± 2.2aA	10.9 ± 1.3aB
	farmlands	73.9 ± 15.4a	4.5 ± 0.6a	–	104.0 ± 15.9aA	1.33 ± 0.05aA	44.5 ± 3.7aB	26.0 ± 3.3aB	15.5 ± 1.7aB	64.8 ± 1.9aA	19.7 ± 0.7aA	11.4 ± 0.6aA
Biocrust	Grasslands	60.5 ± 16.7a	11.6 ± 1.8a	0.13 ± 0.03a	126.7 ± 20.5aA	0.94 ± 0.05bB	64.6 ± 4.9aA	41.5 ± 4.5aA	17.3 ± 1.3aA	66.7 ± 2.1aA	16.0 ± 1.2bB	23.0 ± 0.9aA
	Orchards	47.2 ± 23.7a	7.4 ± 1.2b	0.08 ± 0.01b	114.8 ± 12.3aA	1.15 ± 0.09aB	53.0 ± 5.5bA	33.2 ± 4.2bA	17.2 ± 0.9bA	65.2 ± 1.5bA	17.6 ± 1.6aB	16.7 ± 0.7bA

Note: Lowercase letters denote differences in different land-use type under the same soil crust type at $p = 0.05$ level. Capital letters denote differences in different soil crust type under the same land-use type at $p = 0.05$ level.

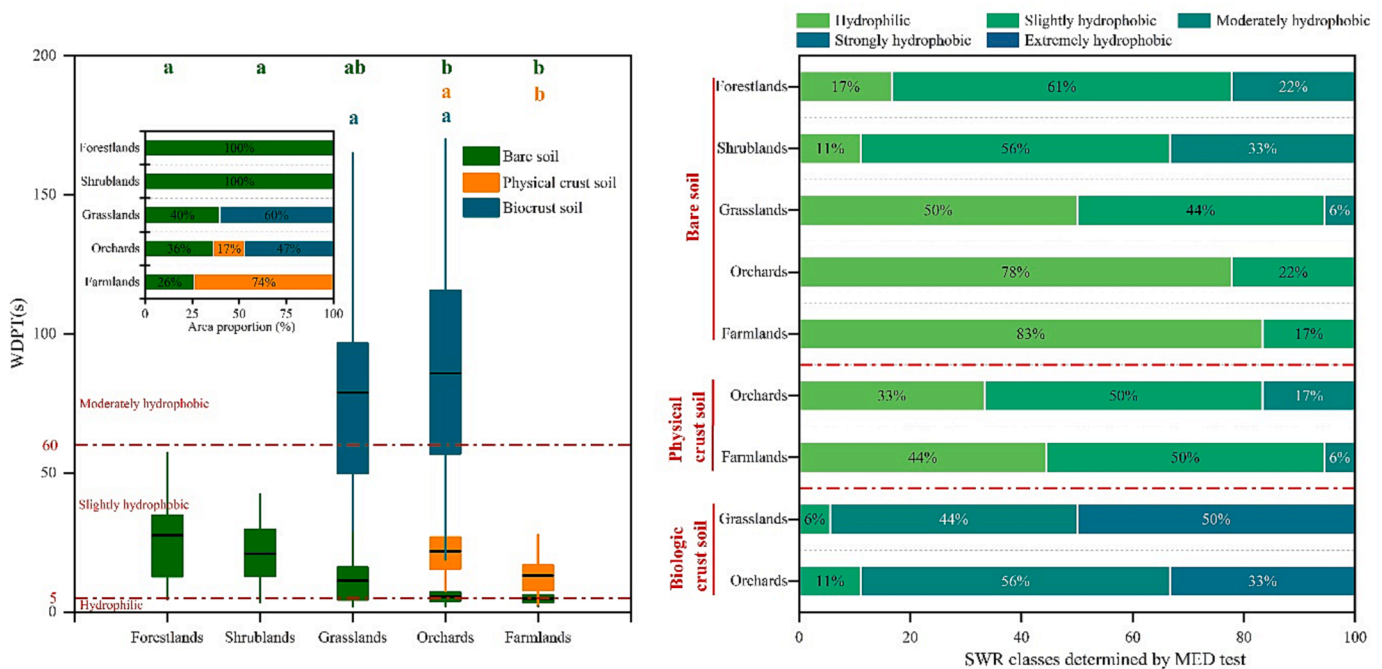


Fig. 3. Soil water repellency measured by the water drop penetration time test (WDPT) and molarity of an ethanol droplet test (MED) under different land-use type and different soil crust. In the box plot, the boxes show the 25th to 75th quantiles, the whiskers are min–max values, the black lines are the mean values. Lowercase letters in the left figure denote the difference between different land-use type under the same soil crust at $p = 0.05$ level.

no significant effect on the soil structure. This may be related to the fact that biocrusts in orchards will be destroyed during weeding and tillage. Many studies have reported that soil crusts significantly increased SWR (Kidron et al., 2020; Li et al., 2022; Sun et al., 2022; Armenise et al., 2018), and the increase in SOM may be an important reason (Armenise et al., 2018; Chandler et al., 2018, Fig. 3, Fig. S4).

4.2. Effects of land-use conversions on soil infiltration

Soil gradually changes from dry to wet with infiltration time, the IIR and SIR can represent the downward flow rate of the soil water under unsaturated and saturated conditions, respectively (Cui et al., 2022b).

Compared with farmlands, revegetation lands had higher infiltration rates, which is consistent with Qiu et al. (2023) and Yang et al. (2017). The relatively small difference in infiltration performance between orchards and farmlands is probably due to the fact that there were no significant differences in either tillage frequency or biomass (above-ground biomass, litter biomass and RMD).

In general, the infiltration performance of the surface layer is higher than that of the subsurface layer due to the more porous structure of the surface soil (Cao et al., 2022; Qiu et al., 2022). However, only the surface IIR of farmlands was higher than subsurface IIR ($p < 0.05$). The surface SIR was not significantly greater than subsurface SIR in the grasslands and orchards. During the initial infiltration stage, the soil is

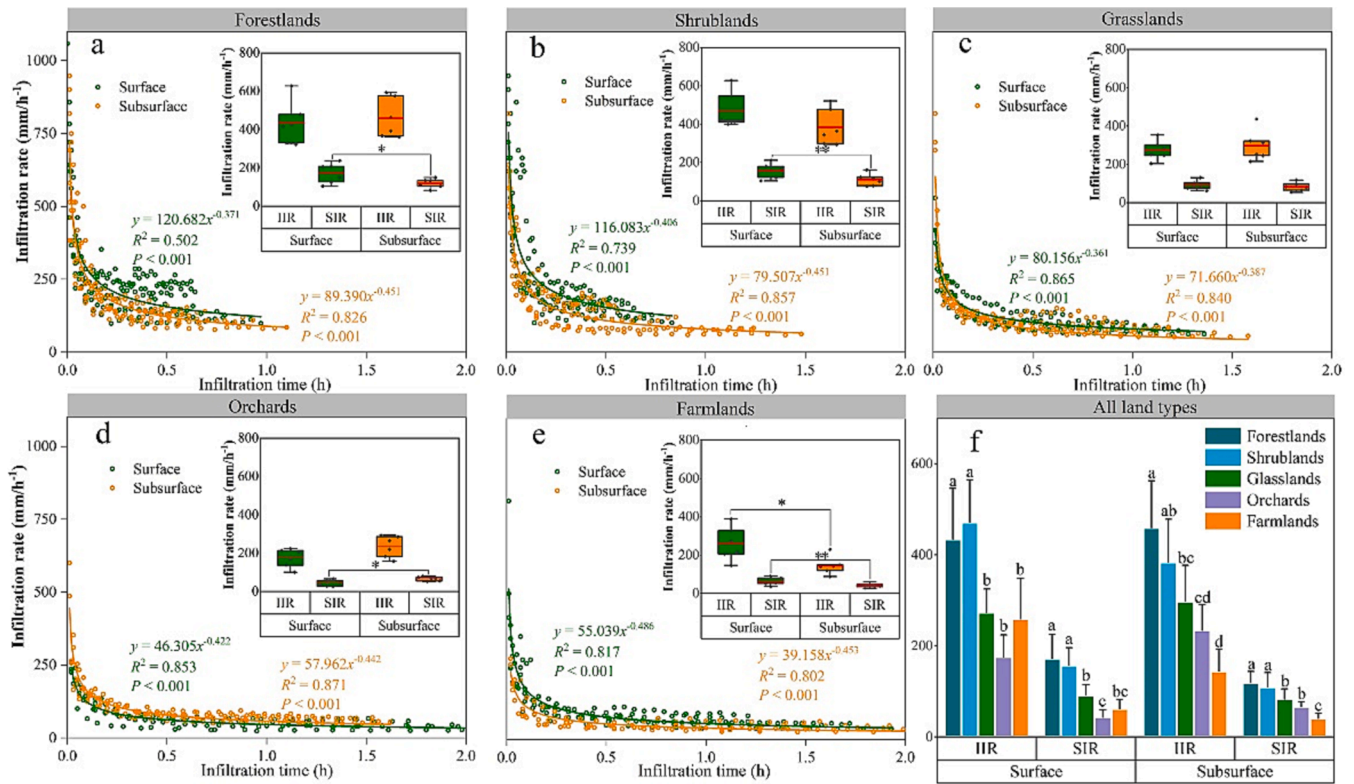


Fig. 4. Change of soil infiltration rates with infiltration time (a-e), initial infiltration rate (IIR) and stable infiltration rate (SIR) of all land types (f). In the box plot, the boxes show the 25th to 75th quantiles, the whiskers are min-max values, the red lines are the mean values. “**” denotes the difference between different soil layers in the same land-use type based on t -test, ** $p < 0.01$, * $p < 0.05$. Lowercase letters in the Figure (f) denote the difference between different land-use type at $p = 0.05$ level. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

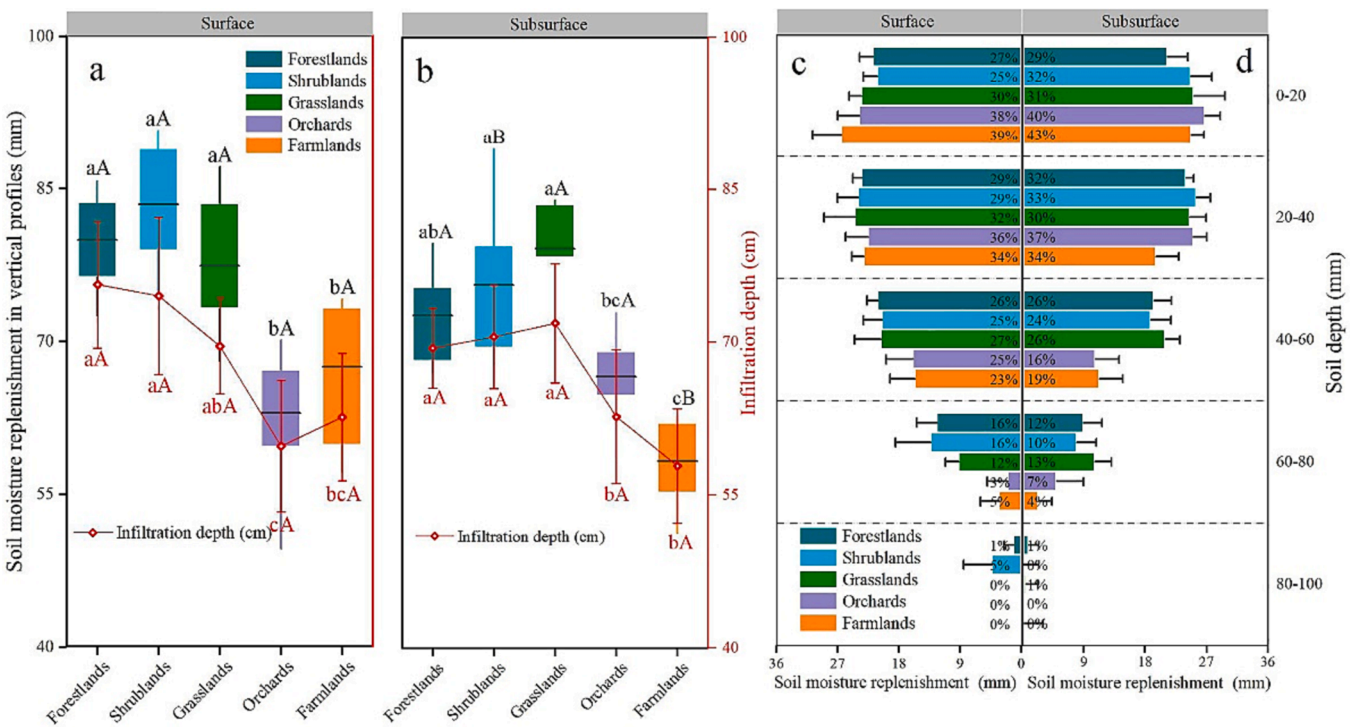


Fig. 5. Soil moisture replenishment and infiltration depth in the vertical profile of the infiltration test conducted in surface (a) and subsurface (b); soil moisture replenishment for each soil layer on the vertical profile of the infiltration test conducted in surface (c) and subsurface (d). In the box plot, the boxes show the 25th to 75th quantiles, the whiskers are min-max values, the black lines are the mean values. Lowercase letters denote differences between different land-use type at $p = 0.05$ level. Capital letters denote differences in tests conducted in different soil layers under the same land-use type at $p = 0.05$ level.

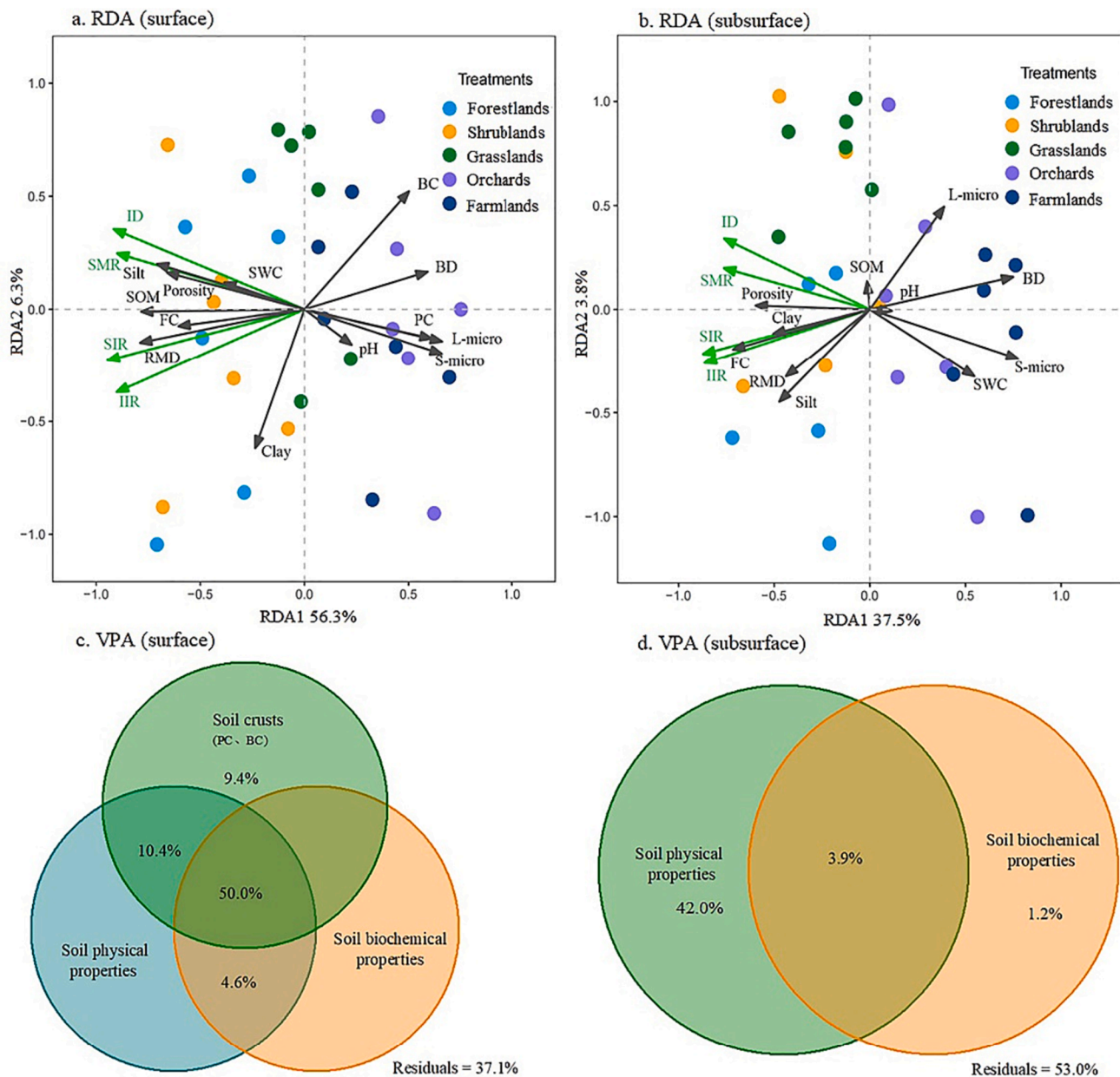


Fig. 6. Redundancy analysis (RDA) and variance partitioning analysis (VPA) in evaluating the effects of soil properties on infiltration properties. Soil properties are shown as black arrows, whereas infiltration properties are presented as red arrows. Values on RDA axes indicate the percentages of total variation explained by each axis. In VPA analysis, soil physical properties include BD, porosity, FC, Macro, L-micro, S-micro, Clay, Silt, Sand and SWC; soil biochemical properties include SOM, Ph, and RMD; soil crusts properties include PC and BC. Percentages are the variations of the infiltration properties explained by soil physical, biochemical and soil crusts properties. IIR: initial infiltration rate; SIR: steady infiltration rate; SMR: soil moisture replenishment; ID: infiltration depth; BD: bulk density; FC: field capacity; Macro: macroaggregate; L-Micro: large microaggregate; S-Micro: small microaggregate; SWC: soil water content; SOM: soil organic carbon; RMD: root mass density; PC: physical crust cover; BC: biocrust cover. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

unsaturated and the presence of preferential flow (Fig. S5) resulting in a higher IIR with high variability in forestlands and shrublands. Moreover, biocrusts in grasslands and orchards may reduce the infiltration rates in the surface (Fig. 8).

The outer ring of the double-ring infiltrometer is designed to ensure that the water in the inner ring infiltrates as evenly and vertically as possible, while the complexity of the infiltration process makes it difficult to avoid the occurrence of preferential flow (Fig. S3) and horizontal flow. Therefore, our measured SMR in the vertical profiles was less than the infiltration amount (Fig. 5). The revegetation lands had higher SMR and ID ($p < 0.05$), meaning that when rainfall occurs, more water will be

able to enter the soil and recharge the deeper soil layers. This is important for arid and semi-arid zones that mainly rely on rainfall as a source of soil water, especially on the Loess Plateau where revegetation has caused dried soil layers due to excessive water use (Jia et al., 2017; Huang and Shao, 2019).

4.3. Influence of soil physicochemical properties and crusts on soil infiltration

Soil physical properties, biochemical properties and crusts were major factors regulating surface infiltration properties, while the

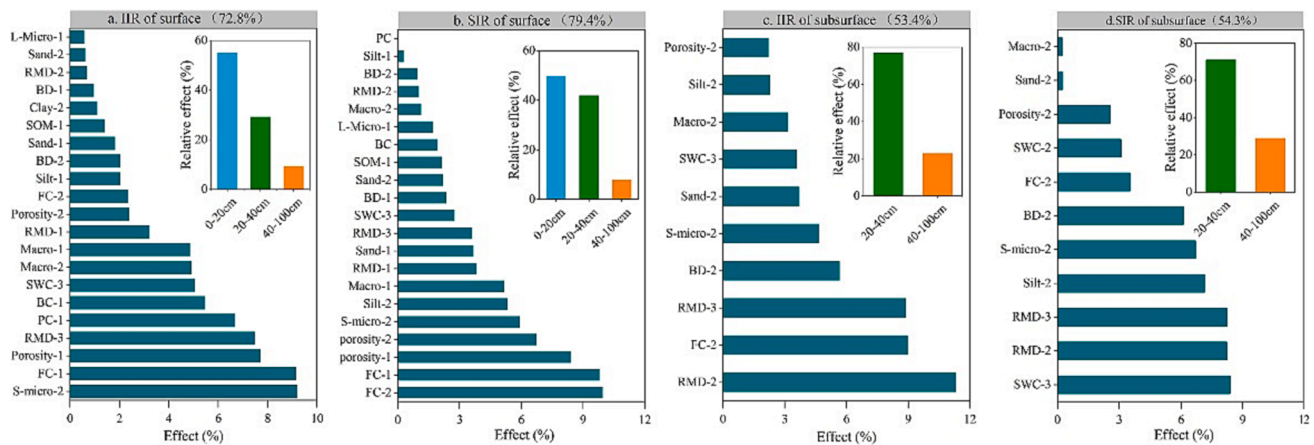


Fig. 7. Effects of soil properties of different soil layers on the initial infiltration rate (IIR) and stable infiltration rate (SIR) of the surface (a, b) and subsurface (c, d), as well as the relative effects of soil properties of different soil layers (inserted figure). BD: bulk density; FC: field capacity; Macro: macroaggregates; L-micro: Large microaggregates; S-micro: Small microaggregates; SWC: soil water content; SOM: soil organic matter; RMD: root mass density; PC: physical crust cover; BC: biocrust cover. The suffixes 1,2,3 after the soil properties indicate 0–20 cm, 20–40 cm and 40–100 cm soil layers respectively.

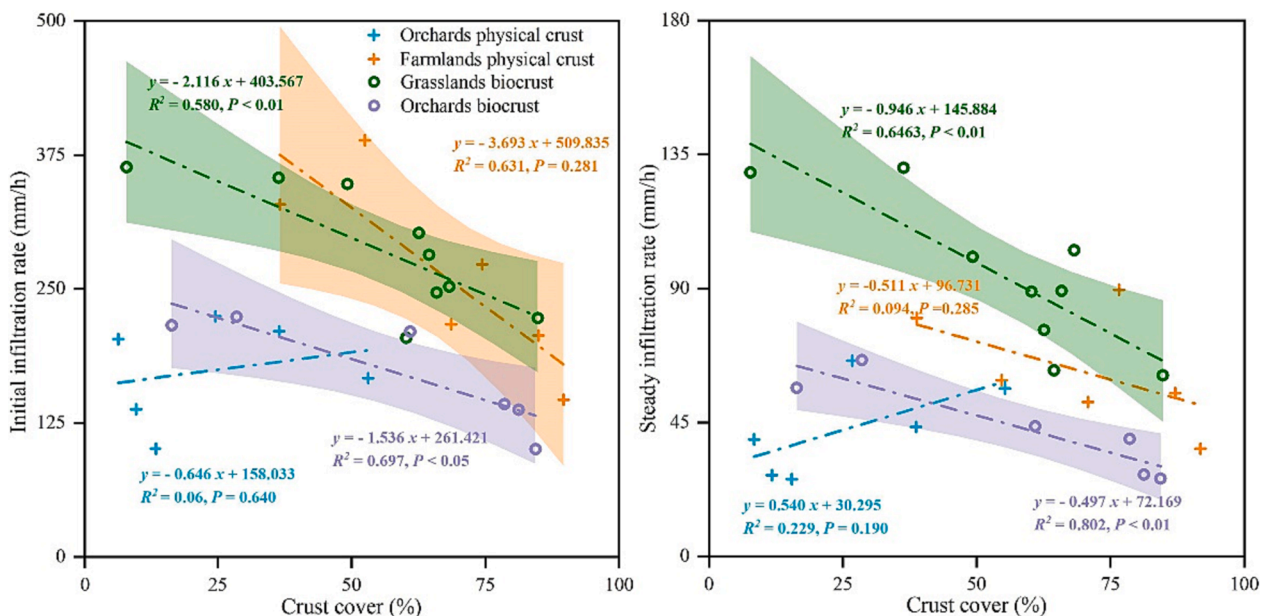


Fig. 8. Regression analysis of infiltration rates and soil crust cover.

subsurface infiltration were mainly influenced by soil physical properties (Fig. 6c-d). Appropriate soil structure can effectively increase the infiltration rates. Soil pores is the main way for water to flow in soil. Numerous studies have shown that lower BD, higher porosity and FC contributed to higher infiltration rates (Beven and Germann, 2013; Jarvis et al., 2013; Zhu et al., 2020). A stable aggregate structure, which often associated with a higher proportion of macroaggregates, will contribute to maintain and stabilize a porous ripening soil layer (Grunwald et al., 2017; Cen et al., 2021). Our study showed that infiltration rates were significantly negatively correlated with L-micro and S-micro (Fig. 6), i.e. significantly positively correlated with macroaggregates, which is consistent with the studies of Sithole et al. (2019) and Zhao et al. (2013). Fine particles (clay and silt) have a higher coefficient of higher viscosity coefficient and soil water suction than sand, and will gradually expand in infiltration process to inhibit water infiltration (Zhao et al., 2022). Some studies found that infiltration rates were significantly negatively correlated with fine particles content (Zhu et al., 2020; Zhao et al., 2022), which is inconsistent with our study (Fig. 6). This may be due to the more dominant promotion of infiltration by soil

structure improvement. In our study, SOM and RMD were significantly and positively correlated with soil infiltration capacity (Fig. 6). SOM could indirectly affect soil infiltration by promoting soil structure and preventing soil consolidation (Wang et al., 2019). Besides, by affecting soil physicochemical properties and the preferential flow paths created by root decay, plant roots can positively influence soil infiltration (Benegas et al., 2014; Wu et al., 2021).

Surprisingly, soil crusts accounted for 69.8% of the variation of infiltration properties in the surface, even if only 9.4% was explained by soil crusts separately (Fig. 6c). Soil biocrust cover in both grasslands and orchards presented a significant negative correlation with IIR and SIR (Fig. 8). However, there was no significant correlation between physical crust cover and infiltration rates in the orchards, probably because of the lower crust cover (16.5%) (Table 4). A significant negative correlation was found between physical crusts and IIR in the farmlands (Fig. 8), while no significant correlation with SIR. This may mean that the inhibition of the infiltration rate by the physical crust was only present in the early stage and gradually disappeared as infiltration progresses. The difference in the effects of physical crusts and biocrusts on infiltration

can be also confirmed by the results of SWR (Fig. 3). SWR can synthesize the influence of various soil properties of soil crusts on water infiltration (Weninger et al., 2019). SWR is a transient soil property (Hermansen et al., 2019) and once sufficiently rewetted, SWR dissipates (Hoogmoed and Stroosnijder, 1984), while when repellent, infiltration will be impeded and runoff will readily generate (Kidron et al., 2022). Generally, the WDPT and MED methods are used to test the persistence and degree of SWR, respectively (Hermansen et al., 2019). Our research showed that compared to physical crusts, biocrusts not only significantly increased the degree of SWR, but also the persistence (Fig. 3).

The impediment of infiltration could be attributed to more fine particles and greater crust thickness (Xiao et al., 2019). Fine particles will gradually expand in infiltration process, blocking the pores subsequently impeding infiltration (Zhao et al., 2022). Due to the thinner crust thickness and lower cohesiveness, the physical crusts may gradually expand and disintegrate during immersion, thus reducing the impeding effect. However, the impeding effect of the biocrusts may present throughout the infiltration process. The biocrusts may be interwoven with the overlying mineral particles into a dense mesh of filaments, sheaths and soil particle, forming a relatively cohesive and (often) a firm layer (Kidron et al., 2022). Besides, although no examined during our research, extracellular polymeric substances (EPS) should not be excluded. Cyanobacteria, bacteria and fungi living on biocrusts excrete large amounts of EPS (Yu et al., 2012), which expand their volume dramatically under conditions of water absorption (by up to 50-fold; see Chenu et al. (1993)).

Driven by gravitational and matrix potential, soil water will move from the surface to the deeper soils. The deeper soils will inevitably affect soil infiltration. However, to the best of our knowledge, few studies have quantified the effect of soils at different depths on infiltration performance. In the Loess Plateau, years of tillage lead to dramatic changes in the properties of the upper soil (0–40 cm), especially the compaction of the subsurface. Then, over two decades of revegetation led to renewed changes in soil properties, which provides an excellent opportunity to study the influence of soils at different depths on water infiltration.

Our study quantified the contribution of different depths of soil to changes in infiltration performance after land-use conversion for the first time (Fig. 7). MLRA showed that the deeper soils had a greater impact on SIR than IIR (Fig. 7). During the initial infiltration stage, the deeper soils was not fully wetted. With increasing infiltration time, more water was transported through the soil pores of deeper soils and the soil was gradually saturated (Cui et al., 2022b). Thus, the influence of the deeper soils on the infiltration rates increased gradually in infiltration process and basically stabilized during the stable infiltration stage. Unexpectedly, for the variation of surface SIR, the relative effect of the 20–40 cm (subsurface) soil properties was 42.1%, even close to that of the 0–20 cm (surface) soil properties (49.8%) (Fig. 7), which was determined by the differed subsurface soil properties among the land-uses. The low hydraulic conductivity caused by the plough pan limited vertical infiltration, thus producing surface runoff, water stagnant (Verbist et al., 2007), and subsurface lateral flow (Wang & Zhang, 2017). In our study, agricultural lands not only had lower infiltration rates than the revegetation lands, but also had lower ID and SMR in the vertical profile of the soil for the same amount of infiltration. This may be due to the plough pan in the agricultural lands resulting in a reduced vertical infiltration capacity, with more water being transported laterally, further contributing to the difference in soil wetness body. In addition, even if some soil properties change slightly, the relative impact of deep soil properties on subsurface IIR and SIR still reached 22.9% and 30.6%. Thus, when assessing and simulating the infiltration capacity of a certain layer of soil, especially the surface soil, it should not only include the soil properties of that layer.

4.4. Implications for land management

Revegetation effectively improved infiltration capacity, but still resulted in a more severe water deficit compared to farmlands (Fig. S3), which may be due to increased evapotranspiration (Jia et al., 2017; Jian et al., 2015). Revegetation has successfully controlled soil erosion in the Loess Plateau, however, it is gradually approaching the limit of sustainable utilization of water resources (Fu et al., 2017). The selection of vegetation restoration species is crucial (Tian et al., 2023), some scholars found that subshrubs and naturally restored grasslands may be the most appropriate species to restore vegetation on the Loess Plateau due to their lower water consumption (Wang et al., 2012; Zhang et al., 2016; Qiu et al., 2022). However, despite the lowest biomass, the grasslands still produced a more severe water deficit than the farmlands, which may be due to biocrusts.

Soil crusts, especially biocrusts, play the role of a disruptor in the infiltration process. The double-ring infiltration test simulates infiltration under ponding conditions, with water being forced to remain in the ring for vertical infiltration. Under natural rainfall conditions, due to the SWR, biocrusts may produce preferential waterflow paths (fingered flow) within the soil profile or surface ponding (Bowker et al., 2013), further reduced rainfall utilization and loss of nutrients. In the revegetation lands, only grasslands showed biocrusts, while other studies reported that biocrusts were also occurred in forestlands (Chen et al., 2020) and shrublands (Liu et al., 2019; Sun et al., 2022). To ensure the sustainability of vegetation restoration, more research is needed on how biocrusts participate in and influence the hydrological cycle.

In agricultural lands, soil crusts also impeded infiltration, even if it could be broken during weeding and tillage. In recent years, some agricultural practices have been reported to help prevent soil crusting and improve water infiltration, for example, straw incorporation (Lin and Chen, 2015), cover cropping (Gabriel et al., 2021) and horticultural fabric mulching (Liao et al., 2021b). Our research showed that the plough pan has a strong influence on soil infiltration of the surface. The plough pan not only restricts the downward transport of water, but also creates barriers to the migration of nutrients and root penetration, consequently hindering crop growth and reducing yield (Zhai et al., 2017; Fenta et al., 2022). Some studies have reported that deep loosening (Olesen and Munkholm, 2007), conservation tillage (Yu et al., 2020) and green manure intercropping (Chen et al., 2019; Fenta et al., 2022) can help to reduce compaction even break up plough pan. Of course, the choice of specific measures depends not only on the effect of soil quality improvement, but also on cost, labor and the effect on yield.

5. Conclusions

Farmland conversions changed soil physicochemical properties and soil crusts, which in turn affected soil infiltration. The revegetation lands had higher soil infiltration capacity due to the loose soil structure, stable aggregates, higher soil organic matter and roots, which allowing more rainwater to enter the soil quickly and transport to deeper layers. Soil crusts impeded infiltration, among which physical crusts impeded infiltration only at the initial stage, while the impeding effect of the biocrusts presented throughout the infiltration process. Soil physical properties, biochemical properties and crusts were major factors affecting surface soil infiltration, while subsurface infiltration was mainly influenced by soil physical properties. Moreover, the relative effect of subsurface soil properties (42.1%) on the steady infiltration rate of the surface was close to that of the surface soil properties (49.8%), which may be due to the plough pan in agricultural lands. Furthermore, more research in future is needed on biocrusts in revegetation lands, which could affect the sustainability of vegetation restoration by hindering infiltration. The measures avoiding the soil crusts and breaking the plough pan in agricultural lands need to be further explored.

CRediT authorship contribution statement

Yang Liao: Conceptualization, Investigation, Data curation, Methodology, Writing – original draft, Writing – review & editing. **Lingbo Dong:** Investigation, Data curation, Methodology. **Ao Li:** Investigation, Data curation. **Wenwen Lv:** Investigation, Data curation. **Jianzhao Wu:** Visualization, Formal analysis. **Hailong Zhang:** Investigation, Data curation. **Ruihua Bai:** Investigation, Data curation. **Yulin Liu:** Methodology. **Jiwei Li:** Methodology. **Zhouping Shangguan:** Writing – review & editing. **Lei Deng:** Conceptualization, Funding acquisition, Supervision, Methodology, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This study was supported by the National Key R&D Program of China (2021YFD1900700), the National Natural Science Foundation of China (U2243225), the Strategic Priority Research Program of the Chinese Academy of Sciences (XDA23070201).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2023.130283>.

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